

Radix Sort Algorithm : $\rightarrow$ [Brother of Count Sort]

- * Multi-digit numbers
- * Constant length Strings
"radar" "brave" "small"
- * Non-comparison algorithm
- * It is inspired from "Bucket Sort"

Radix Sort : 1's, 10's, 100's & so on. Count Sort ()

Steps :

input

325 | 009 | 256 | 072 | 003

1. Max : 325

2. Digits in max = 3

So, no of passes = 3

3. Create 0-9

10 Buckets

4. Add zeros
to balance.
(theory)

072 | 003 | 325 | 256 | 009 | I1

003 | 009 | 325 | 256 | 072 | I2

003 | 009 | 072 | 256 | 325 | I3

0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

$\rightarrow$ 072 003 325 256 009

003
009
003, 009,
072

325
256 325

$O(n+k)$

max = 325

exp = 1, $1 \times 10 = 10$, $10 \times 10 = 100$,
 $100 \times 10 = 1000$

$$\textcircled{1} \text{ max/exp} = 325/\text{exp} = 325/1 = 325 \% 10 = 5$$

$$\textcircled{2} \text{ max/exp} = 325/10 = 32 \% 10 = 2$$

$$\textcircled{3} \text{ max/exp} = 325/100 = 3 \% 10 = 3$$

$$\text{stop max/exp} = 325/1000 = 0 \rightarrow \text{stop}$$

```
for (int exp = 1; max/exp > 0; exp *= 10)
{
    countSort(arr, size, exp);
}
```